GH RAISONI COLLEGE • DEPARTMENT OF BBA-CA • PUNE CAMPUS

AI CHATBOT FOR COLLEGE — REVISED SYSTEM SPECIFICATIONS

Explicit Database Swimlane showing where student records are stored and fetched

1. Activity Diagram (With Dedicated Database Swimlane)

✔ Faculty Action Item: Explicit Database Representation in Activity Flow

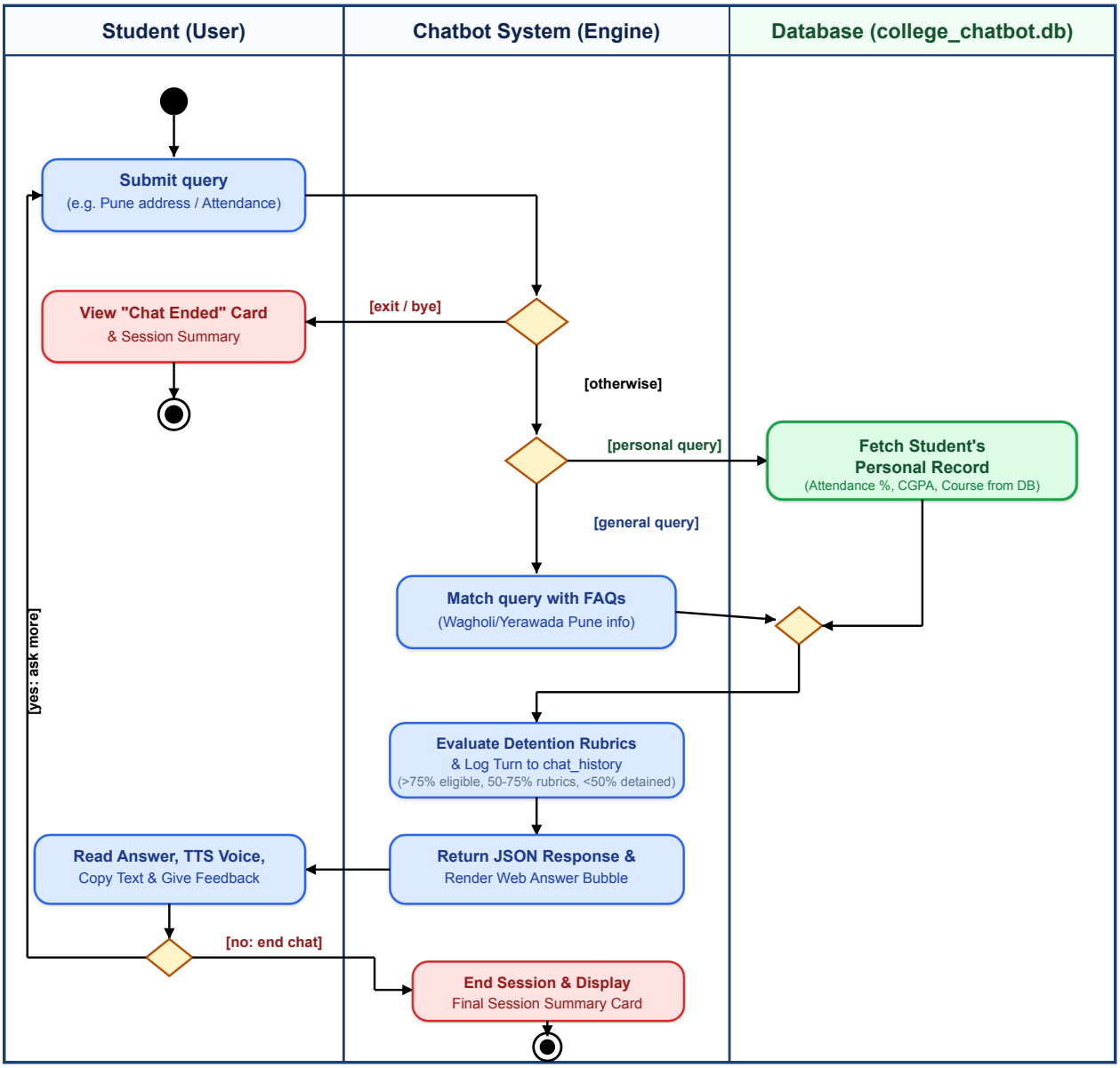


Fig 1: Activity Diagram — 3-Swimlane architecture (Student, Chatbot System, Database) showing explicit personal record fetching and chat logging.

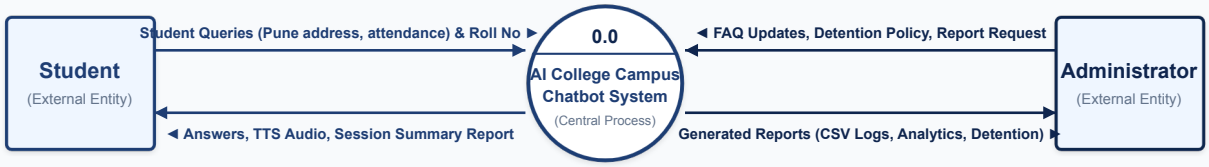
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First-Level data flow decomposition with external entities, processes, data stores, and symbol guide

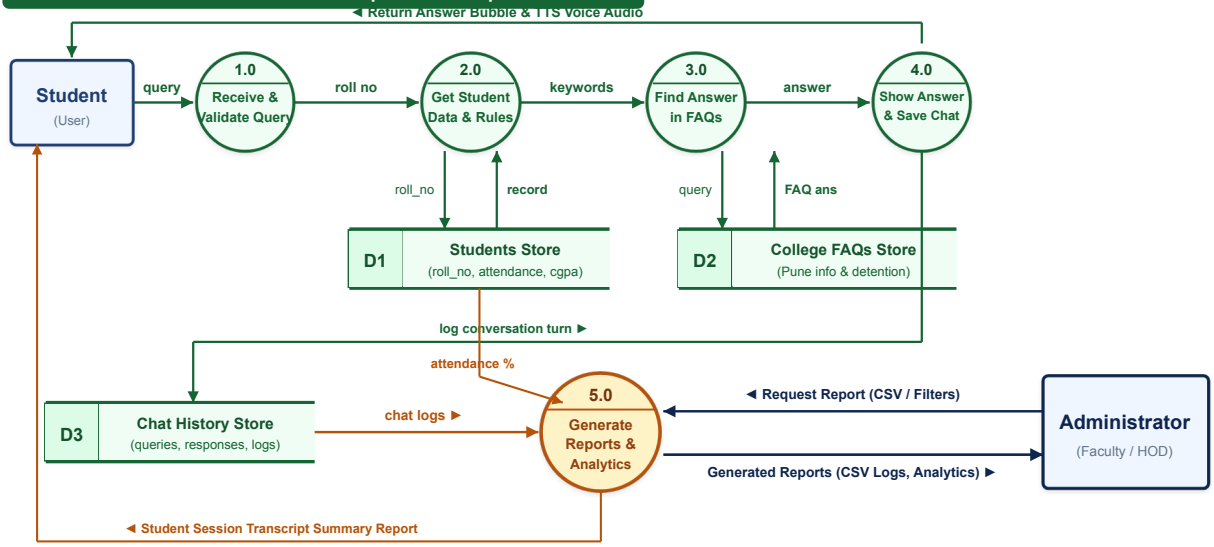
2. Data Flow Diagram (DFD Level 0 Context & Level 1 Process Breakdown with Report Generation)

✔ Faculty Action Item: First-Level DFD & Complete Symbols Table

DFD Level 0: Context Diagram



DFD Level 1: First-Level Process Decomposition & Report Generation



Standard DFD Notation & Symbol Conventions (Gane-Sarson & DeMarco)

Symbol Shape	DFD Element	Standard Academic Definition	Implementation in Our Project
▭	External Entity	Origin/destination of data outside boundary (Square)	**Student (User) & Administrator (Faculty/HOD)**
○	Process (Bubble)	Transforms input data into output data (Circle with ID) Calculates, retrieves, formats, or generates reports	**1.0 to 4.0 Query Handlers & 5.0 Generate Reports** Query answering & administrative CSV/analytics reporting
D |	Data Store *(Open-Ended)*	Repository of data at rest (Open parallel lines) **MANDATORY RULE: Right side must be open; NEVER closed**	**D1 Students, D2 College FAQs, D3 Chat History** Persistent SQLite tables feeding search & report generation
→	Data Flow	Directed solid arrow showing data path & payload Transfers packets between entities, processes, and stores	**query, record, FAQ answer, generated reports CSV** Directional vectors with explicit noun-phrase labels

Fig 2: Data Flow Diagram (DFD) — Properly formatted Level 0 Context Diagram (System boundary with Student & Administrator) plus Level 1 Process Decomposition (1.0 to 4.0 Query Handlers & 5.0 Generate Reports & Analytics) with standard open-ended data stores (D1 Students, D2 FAQs, D3 Chat History) and formal academic DFD symbols reference table.

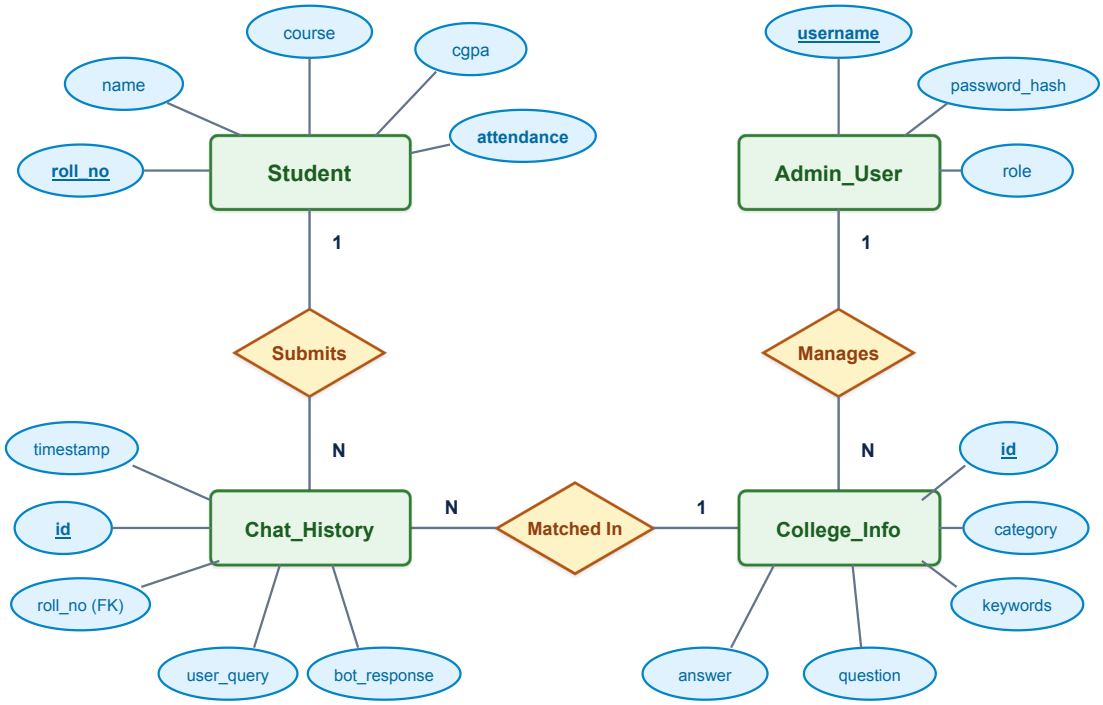
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Classic ER modeling with 4 core entities, attributes, cardinalities, and symbol legend

3. Entity-Relationship (ER) Diagram — Classic Notation

✔ Faculty Action Item: Classic ER Notation matching Faculty Specification

ER Diagram (Classic Notation): AI Chatbot for College



Schema & Project Knowledge Base Defense:

- **Student Entity:** Stores attendance %, CGPA & evaluates 4-tier detention policy.
- **College_Info Entity:** Contains Pune campuses (Wagholi & Yerawada) & exam rules.
- **Chat_History Entity:** Logs each turn with student roll_no FK, query & bot answer.
- **Admin_User Entity:** RBAC authentication for updating college FAQs & detention data.

Verified in 3NF Normalization: Zero transitive dependencies & lossless join preservation.

Key / Legend:

- Entity (Table)
- Attribute (Column) — id = Primary Key
- Relationship
- Link / Cardinality (1:N)

Fig 3: Entity-Relationship Diagram (ERD) — Classic notation with Student, Admin_User, Chat_History, and College_Info.

AI CHATBOT FOR COLLEGE — REVISED SYSTEM SPECIFICATIONS

Detailed structural justification, attribute breakdown, normalization, and relational flow

4. Entity Review & Justification (3NF Schema Analysis)

✔ Faculty Action Item: Formal Entity Review & Relational Justification

ENTITY REVIEW & JUSTIFICATION MANIFESTO (3NF SCHEMA ARCHITECTURE)

Comprehensive analysis of the 4 core entities, structure, attributes, normalization, and relational flow as requested in faculty action items.

Entity Name	Primary Key	Foreign Key(s)	Key Attributes	System Role & Justification
Student	`roll_no`	None (Base Entity)	name, course, cgpa, attendance	Represents enrolled students. Enables personalized query responses regarding attendance percentage, detention risk, and academic standing.
Admin_User	`username`	None (Independent)	password_hash, role	Stores administrative credentials with salted SHA-256 hashes. Restricts knowledgebase CRUD operations and audit log exports to authorized staff.
College_Info	`id`	None	category, keywords, question, answer	The institutional knowledge base. Contains categorized FAQ entries (Admissions, Fees, Exams, Pune Campus Location, Detention Rubrics) matched by the NLP engine.
Chat_History	`id`	`roll_no` (FK)	user_query, bot_response, timestamp	Audit telemetry and analytics log. Records every student interaction, resolved FAQ answer, timestamp, and feedback for administrative review and training.

Relational Cardinality & Integrity Rules:

- **Student (1) — Submits — (N) Chat_History:** A registered student can submit multiple inquiries over time. If a query is submitted anonymously by a guest, roll_no is set to NULL (optional 1:N).
- **College_Info (1) — Matched In — (N) Chat_History:** A specific FAQ record can match multiple historical student questions across sessions.
- **Admin_User (1) — Manages — (N) College_Info:** Authorized department administrators create and update knowledgebase records.

Why Exactly These 4 Entities? (Normalization Defense)

- **Eliminates Redundancy:** Separating student personal data from chatbot logs prevents duplicating student names/CGPA in every chat record.
- **Third Normal Form (3NF):** Every non-key attribute depends strictly on the primary key, the whole key, and nothing but the key.
- **Security Isolation:** Administrator credentials remain physically and logically isolated from public FAQ and student interaction data.

Fig 4: Entity Review & Justification Table — Formal 3NF relational breakdown of the 4 core database entities.

AI CHATBOT FOR COLLEGE — REVISED SYSTEM SPECIFICATIONS

Official Theory and Practical Detention Rubrics trained into chatbot knowledge base

5. Implementation of Student Detention Policy

✔ Faculty Action Item: Official Detention Policy Integration & Rubrics

OFFICIAL STUDENT DETENTION POLICY & CHATBOT LOGIC IMPLEMENTATION

Trained into Chatbot NLP Knowledge Base and Personalized Response Logic as clarified in Faculty Meeting.

1. THEORY DETENTION RUBRICS

(Attendance Excluding Medical Grounds)

S.No	Attendance Bracket	Academic Rubrics & Requirements	Eligibility Status
—	Above 75.0%	Regular academic standing. No additional rubrics required.	✔ Fully Eligible
1	60.0% to 75.0%	End Sem Examination Paper Sets of previous 2 years to be solved in **Open Book mode** in the College.	Eligible after solving sets
2	50.0% to 59.99%	(1) End Sem Examination Paper Sets of previous 2 years solved in Open Book mode in College. (2) Must appear for a prelim examination and score a **minimum of 60% marks** to get out of detention.	Eligible if prelim ≥ 60%
3	Below 50.0%	**No consideration. Strictly detained from university end-semester examinations.**	✘ Strictly Detained

2. PRACTICAL DETENTION RUBRICS

(Note: Not more than 2 practicals to be conducted in a day)

S.No	Attendance Bracket	Practical Rubrics & Penalties	Practical Eligibility
1	60.0% to 75.0%	The student will **not be eligible for the attendance mark component** of internal practical marks.	Attendance marks forfeited
2	50.0% to 59.99%	Student must appear for a **practical-based prelim examination** and score a minimum of 60% marks to be eligible to get out of detention.	Eligible if practical ≥ 60%
3	Below 50.0%	**No consideration. Strictly detained from practical examinations.**	✘ Strictly Detained

Chatbot Engine Integration Details:

When a logged-in student asks *"What is my attendance?"*, app.py evaluates their percentage against these exact 4 thresholds and returns their specific status, eligible exam status, and designated academic rubrics. Both Pune Campus addresses — Wagholi Campus (Gat 1200, Domkhel Rd, Pune - 412207) and Yerawada Campus (Nyati Unitree, Airport Rd, Yerawada, Pune - 411006) — are active in institutional FAQs.

Fig 5: Implementation of Student Detention Policy — Official Theory and Practical rubrics table trained into chatbot engine.